Solution

PVK Mock Exam Analysis I: One Variable

Surname XX

First Name XX

Legi-No. **XX-456-341**

Exam-No. *234*

Disclaimer: This is a Mock Exam designed for Exam Preparation and not an official Exam!

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Please do not turn the page!

Multiple Choice Questions

There is exactly one correct answer for each of the following questions. No negative points are given.

MC1 [**2 Points**]
Find the minimum on $(0, \infty)$ of the function

$$f(x) = x + \frac{4}{x}.$$

(A) 2

(B) 3

(C) 4

(D) 5

Solution: C

MC2 [**2 Points**]
For which $\alpha \in \mathbb{R}$ does the series

$$\sum_{n=2}^{\infty} \exp\left(n(\alpha \log(n) - \log(2^n + n^2))\right)$$

converge?

(A) $\alpha > 1$

(B) $\alpha \in \mathbb{R}$

(C) $\alpha \geq 1$

(D) $\alpha < 1$

Solution: B

MC3 [**2 Points**]
Compute the radius of convergence of the power series

$$\sum_{n=1}^{\infty} \left(1 + \frac{(-1)^n}{n}\right)^{n^2} x^n.$$

(A) $\frac{1}{e}$

(B) 1

(C) 0

(D) e

Solution: A

MC4 [**2 Points**]
Compute the limit

$$\lim_{x\to 0} \frac{e^{2x} - 1 - 2x}{x^2}.$$

(A) 0

(B) 1

(C) 2

(D) 4

Solution: C

MC5 [**2 Points**]
Compute the integral

$$\int_0^1 \sqrt{1 - x^2}\, dx.$$

(A) $\frac{\pi}{2}$

(B) $\frac{\pi}{4}$

(C) 1

(D) 2

Solution: B

MC6 [**2 Points**]
Consider the first order ODE

$$u'(x) = \frac{2}{x}u(x) + x, \qquad u(1) = 0.$$

Find $u(2)$.

(A) $4\log(2)$

(B) $2\log(2)$

(C) 2

(D) $\log(2)$

Solution: A

MC7 [**2 Points**]
Consider the second order ODE

$$u'' - u = e^x, \qquad u(0) = 0,\ u'(0) = 1.$$

Find $u(1)$.

(A) $\frac{3e^2-1}{4e}$

(B) $\frac{3e}{4}$

(C) $\frac{e}{4}$

(D) $\frac{3}{4e}$

Solution: A

MC8 [**2 Points**]
Let $(x_n)_{n\geq 1}$ be a sequence.

(A) Every monotone sequence is bounded.

(B) Every bounded sequence has a convergent subsequence.

(C) Every Cauchy sequence is monotone.

(D) Every divergent sequence is unbounded.

Solution: B

MC9 [**2 Points**]
Which improper integral converges?

(A) $\displaystyle\int_1^\infty \frac{dx}{x}$

(B) $\displaystyle\int_1^\infty \frac{1}{x^{3/2}}\,dx$

(C) $\displaystyle\int_0^1 \frac{dx}{x}$

(D) $\displaystyle\int_0^\infty e^x\,dx$

Solution: B

MC10 [**2 Points**]
Which sequence diverges?

(A) $\sqrt{n+1}-\sqrt{n}$

(B) $\frac{\log n}{n}$

(C) $\sin\left(\frac{1}{n}\right)$

(D) $\frac{\log(n)}{\log(\log(n))}$

Solution: D

MC11 [**2 Points**]
Which set is closed in $\mathbb{R}$?

(A) $(0, 1)$

(B) $[0, 1)$

(C) $\{1/n : n \in \mathbb{N}\}$

(D) $[1, \infty)$

Solution: D

MC12 [2 Points]
Let $f : \mathbb{R} \to \mathbb{R}$. Which identity always holds?

(A) $f(A \cap B) = f(A) \cap f(B)$

(B) $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$

(C) $f(f^{-1}(A)) = A$

(D) $f^{-1}(f(A)) = A$

Solution: B

Box Questions

Only the final answer counts! No partial credit can be given.

K1 [**2 Points**]
Compute the indefinite integral

$$\int \frac{3x-2}{x(x+1)}\,dx.$$

Solution: Perform partial fraction decomposition:

$$\frac{3x-2}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1}.$$

Then $3x-2 = A(x+1)+Bx = (A+B)x + A$. Comparing coefficients: $A+B=3$, $A=-2$. Thus $B=5$. Hence

$$\int \frac{3x-2}{x(x+1)}\,dx = \int \left(\frac{-2}{x} + \frac{5}{x+1}\right) dx = -2\ln|x| + 5\ln|x+1| + C.$$

K2 [**2 Points**]
Compute the indefinite integral

$$\int \frac{\cos x}{1+\sin x}\,dx.$$

Solution: Let $u = 1+\sin x$, then $du = \cos x\,dx$.

$$\int \frac{\cos x}{1+\sin x}\,dx = \int \frac{1}{u}\,du = \ln|u| + C = \ln|1+\sin x| + C.$$

K3 [**2 Points**]
Compute the Taylor expansion at 0, up to order 4, of

$$e^{x^2} - \cos x.$$

Solution: Recall: $e^{x^2} = 1 + x^2 + \frac{x^4}{2} + o(x^4)$, $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4)$. Thus

$$e^{x^2} - \cos x = \left(1 + x^2 + \frac{x^4}{2}\right) - \left(1 - \frac{x^2}{2} + \frac{x^4}{24}\right) + o(x^4) = \frac{3}{2}x^2 + \frac{11}{24}x^4 + o(x^4).$$

Proofs from the Lecture

The whole solution is being graded. Partial credit can be given.

B1 [**7 Points**]
Assume that a sequence of Riemann integrable functions $f_n : [a, b] \to \mathbb{R}$ converge uniformly $f_n \to f$. Show that f is integrable and

$$\lim_{n\to\infty} \int_a^b f_n(x)dx = \int_a^b f(x)dx.$$

See Lecture Notes.

B2 [**8 Points**]
Show that every continuous function $f : [a, b] \to \mathbb{R}$ is integrable.

See Lecture Notes.

Application of the Theory

The subtasks can be solved independently from another. Partial credit can be given.

A1 [9 Points]
Let $f : [0, \infty) \to \mathbb{R}$ be a continuously differentiable function such that

$$\lim_{x\to\infty} f(x) = 0 \quad \text{and} \quad \int_0^\infty |f'(x)|\, dx < \infty.$$

(A) Prove that f is bounded on $[0, \infty)$.

(B) Show that for all $x \geq 0$,

$$|f(x)| \leq \int_x^\infty |f'(s)|\, ds. \tag{$*$}$$

(C) Deduce from the integral inequality $(*)$ in (B) that $f(x) \to 0$ uniformly as $R \to \infty$, i.e. for every $\varepsilon > 0$ there exists $R > 0$ such that

$$|f(x)| < \varepsilon \quad \forall x \geq R.$$

Solution:

(A) Since $\lim_{x\to\infty} f(x) = 0$, there exists $R > 0$ such that

$$|f(x)| \leq 1 \quad \forall x \geq R.$$

By continuity, f is bounded on the compact interval $[0, R]$. Hence f is bounded on $[0, \infty)$.

(B) We get by the Fundamental Theorem of Calculus:

$$|f(x)| = |f(x) - \underbrace{\lim_{x\to\infty} f(x)}_{=0}| = \left|\int_x^\infty f'(s)\, ds\right| \leq \int_x^\infty |f'(s)|\, ds.$$

(C) We know by (B) that

$$|f(x)| \leq \int_x^\infty |f'(s)|\, ds.$$

We know that as $R \to \infty$ the integral

$$\int_R^\infty |f'(s)|\, ds \to 0.$$

Therefore we can choose $R > 0$ big enough so that the integral is smaller than an arbitrary $\varepsilon > 0$. For all $x \geq R$ we therefore get the desired result:

$$|f(x)| \leq \int_x^\infty |f'(s)|\, ds \leq \int_R^\infty |f'(s)|\, ds < \varepsilon.$$

A2 [11 Points]
Let $f : [0, 1] \to \mathbb{R}$ be a *non-constant* and *twice continuously differentiable* function such that

$$f(0) = f(1) = 0 \quad \text{and} \quad \int_0^1 f(x)\, dx = 0.$$

(A) Show that there always exists a value $\xi \in [a, b]$ such that

$$\int_a^b f(x)dx = f(\xi)(b-a).$$

Hint: Consider the continuous function

$$g(t) := \int_a^t f(x)\,dx - \frac{t-a}{b-a}\int_a^b f(x)\,dx$$

and use Rolle's Theorem.

(B) Deduce from (A), that there exists $x_0 \in (0, 1)$ such that $f(x_0) = 0$.

(C) Prove that there exist points $a, b \in (0, 1)$ with

$$f'(a) > 0 \quad \text{and} \quad f'(b) < 0.$$

(D) Show that there exists $c \in (0, 1)$ such that

$$f''(c) < 0.$$

Solution:

(A) Define

$$F(t) := \int_a^t f(x)\,dx \quad \text{for } t \in [a, b].$$

Then F is continuously differentiable on $[a, b]$ with

$$F'(t) = f(t).$$

Consider the continuous function

$$g(t) := F(t) - \frac{t-a}{b-a}\int_a^b f(x)\,dx.$$

We have

$$g(a) = 0 \quad \text{and} \quad g(b) = 0.$$

By Rolle's theorem, there exists $\xi \in (a, b)$ such that $g'(\xi) = 0$, hence

$$f(\xi) - \frac{1}{b-a}\int_a^b f(x)\,dx = 0.$$

Therefore,

$$\int_a^b f(x)\,dx = f(\xi)(b-a).$$

(B) Apply part (A) to the interval $[0, 1]$. Since

$$\int_0^1 f(x)\,dx = 0,$$

there exists $x_0 \in (0, 1)$ such that

$$0 = f(x_0)(1-0).$$

Hence $f(x_0) = 0$. Together with $f(0) = f(1) = 0$, this shows that f has at least three zeros in $[0, 1]$, and in particular one in $(0, 1)$.

(C) Since f is non-constant and continuous on $[0,1]$ with $f(0) = f(1) = 0$, it must attain either a positive maximum or a negative minimum at some point in $(0,1)$. Assume without loss of generality that f attains a positive maximum at $x_{\max} \in (0,1)$. Then $f'(x_{\max}) = 0$.

Because $f(0) = 0 < f(x_{\max})$, the Mean Value Theorem implies the existence of $a \in (0, x_{\max})$ such that

$$f'(a) > 0.$$

Similarly, since $f(x_{\max}) > 0 = f(1)$, there exists $b \in (x_{\max}, 1)$ such that

$$f'(b) < 0.$$

(D) From part (C), choose $a < b$ in $(0,1)$ such that

$$f'(a) > 0 \quad \text{and} \quad f'(b) < 0.$$

The function f' is continuously differentiable, hence continuous, on $[a,b]$. By the Mean Value Theorem applied to f' on $[a,b]$, there exists $c \in (a,b)$ such that

$$f''(c) = \frac{f'(b) - f'(a)}{b - a}.$$

Since $f'(b) - f'(a) < 0$, it follows that

$$f''(c) < 0.$$

Approximate Grade Scheme

Total: 65 Points
6.0: approx. 90%
5.75 : approx. 85%
5.5: approx. 80%
5.25: approx. 75 %
5.0: approx. 70%
4.75: approx. 65%
4.5: approx. 60%
4.25: approx. 55%
4.00: approx. 50%
3.75: approx. 45%
3.5: approx. 42%
3.25: approx. 37.5%
3.0: approx. 33%
2.75: approx. 28%
2.5: approx. 25%
2.25: approx. 20%
2.0: approx. 16%
1.75: approx. 11%
1.5: approx. 8%
1.25: approx. 3%
1.0: approx. 0%